Simulation-Driven Bimanual Garment Folding

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Abstract

Robotic cloth manipulation remains one of the most challenging open problems in robot learning due to the high-dimensional state space of deformable objects, partial observability caused by self-occlusions, and complex nonlinear dynamics that resist analytical modeling. In this work, we present an entry to the LeHome Challenge 2026 that trains a generalizable bimanual clothes-folding policy across four clothing categories in a simulated home environment. Our approach fine-tunes SmolVLA, a vision–language–action model, through a training pipeline consisting of supervised fine-tuning on teleoperated demonstrations. To address the limited simulation data provided by the challenge, we employ NVIDIA Cosmos-Transfer 2.5 to generate photorealistic augmented data and improve policy robustness. We further benchmark our method against two widely used imitation learning baselines: Action Chunking Transformer (ACT) and Diffusion Policy. The experimental results show that the resulting SmolVLA policy achieves superior performance in simulation, reaching a task success rate of 69% on preliminary evaluation and outperforming both baselines. Our contributions include a trained policy for the challenge environment, a formal challenge submission, and an empirical analysis of the viability of SOTA robotic policy architectures for complex deformable object manipulation tasks such as garment folding.

1 Introduction

Robotic garment manipulation remains a significant challenge in the field of autonomous control due to the unique physical properties of fabric. Unlike rigid objects, cloth is continuously deformable, possessing an extremely high-dimensional state space with highly nonlinear dynamics that are exceptionally difficult to model analytically. Moreover, the task of folding introduces the challenge of severe partial observability. As a garment is folded over itself, significant portions of the fabric surface become self-occluded and unobservable from any camera configuration. Finally, to be practically viable, robotic manipulation policies must achieve a high degree of generalization. A robust policy cannot simply overfit to a single trajectory; it must dynamically adapt to unseen garment styles, varying textures, and workspace environments.

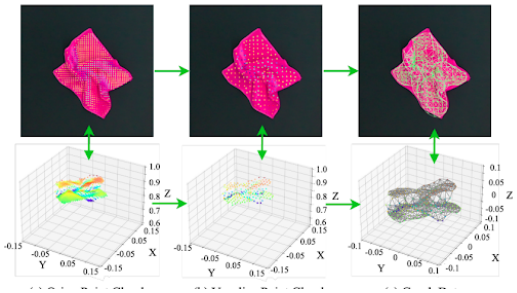

Figure 1: Cloth configuration space

2 System Overview and Methodology

Our system is implemented within the LeHome simulation environment, utilizing the LeRobot framework for policy training and deployment. The garment folding task is defined as a goal-conditioned bimanual manipulation problem across four distinct clothing categories.

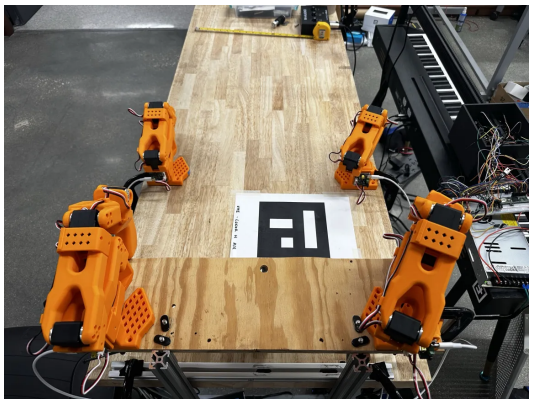

Figure 2: SO101 Leader-Follower Teleoperation Setup used for expert data collection.

2.1 Action and Observation Space

The hardware configuration utilizes a bimanual setup consisting of two LeRobot SO-ARM101 robotic arms as shown (Fig 1.), totaling 12 Degrees of Freedom (DoF) (6 DoF per arm). Actions are commanded as joint positions in a continuous space, enabling the coordinated movements required to handle thin-shell deformable objects.

The observation space $\mathcal{O}$ is strictly vision-based and comprises three synchronized RGB camera streams:

- **Workspace Camera:** A static, top-down view providing global context for garment localization and orientation.
- **Wrist Cameras:** Two end-effector-mounted cameras (Left and Right) provide local feedback for precise grasping and alignment of fabric edges.

2.2 Data Collection and Augmentation

To teach the robot how to fold clothes, the first thing we had to do was show it what a successful fold actually looks like. We built a teleoperation setup where a human operator controls a LeRobot SO-ARM101 "leader" arm. By physically guiding this arm, the human essentially walks the robot through the entire folding process step by step. This hands-on method lets us capture demonstrations that are clean and precise, including the small, careful movements needed when dealing with soft materials like cloth.

However, we didn't want the robot to simply memorize one perfect scenario. In real life, clothes rarely start out neatly placed on a table. So while collecting demonstrations, we deliberately created messy starting conditions. We crumpled the clothes, twisted them, and dropped them onto the table from different angles and positions. This forced the model to learn how to recognize and recover from cluttered, unpredictable situations instead of relying on ideal setups.

Another big challenge in robotics is making sure the system still works when the environment changes. A robot trained in one room might struggle in another because of different lighting, backgrounds, or table surfaces. Instead of collecting hundreds of hours of new real-world data, we built a data-augmentation pipeline using NVIDIA Cosmos-Transfer 2.5. This tool takes our original demonstrations and automatically generates many realistic variations of them.

In these generated videos, the backgrounds change, lighting conditions shift, colors and textures of clothes and tables vary, and random objects appear in the scene. This process—known as domain randomization—greatly expands the diversity of training data. By learning from these varied examples, the robot's vision system learns to ignore irrelevant details in the environment and focus on what actually matters: the cloth and the folding task itself.

2.3 Policy Architectures

We compared three distinct architectures for imitation learning currently popular in the field of robotic manipulation. A general introduction of each is below.

2.3.1 Diffusion Policy

Diffusion Policy frames action prediction as a denoising diffusion process (similar to diffusion models for image generation), iteratively refining an action sequence from randomly initialized noise into a clean output conditioned on sensory observations (2). Like ACT, diffusion policy implementations (including ours) predict a sequence of 'k' action steps rather than a single output, encouraging long-term planning. This learned denoising function is built upon a Transformer backbone, and unlike ACT excels at representing multimodal action distributions—making it well-suited for tasks where multiple valid solution trajectories exist. However, it requires executing a fixed number of denoising steps at inference, meaning that inference is comparatively resource-intensive and slow.

2.3.2 Action Chunking Transformer (ACT)

ACT takes a more direct approach than diffusion policy, directly predicting a short horizon of future actions (or "chunk") in a single forward pass (3). This chunking strategy reduces compounding errors and improves temporal consistency. Compared to diffusion policy, ACT is data-efficient, achieving strong performance even from relatively small demonstration datasets and with fewer weights, with lower latency at inference time. However, ACT struggles in situations where multiple valid solution trajectories exist.

2.3.3 Vision-Language Action Model (SmolVLA)

VLA represents a newer class of policy that integrates a pre-trained vision-language model with

an action head which, similar to the above, provide control outputs to the robot. We based our implementation on SmolVLA, HuggingFace's lightweight VLA trained on open-source robotics and video data (4). By grounding policy learning in representations already trained on large-scale vision-language data, SmolVLA is designed to generalize across tasks and environments with minimal task-specific fine-tuning. This architecture prioritizes cross-task generalization and transfer, making it especially relevant in settings where data collection is expensive or where zero-shot and few-shot performance is desirable; however, it may struggle even under fine-tuning in tasks that are out-of-distribution for its original training data (e.g. highly specialized industrial tasks for which open source video may not have been available).

3 Experimental Results

Evaluation is conducted in Isaac Sim using the proximity of hardcoded keypoints in the garment meshes. A garment is considered to have been successfully folded when each of these hardcoded keypoint pairs are within a certain distance threshold of each other (or further than a threshold in the case of keypoints that should be separated in the folded state) (7). This metric is commonly used in cloth manipulation research because of its simplicity to evaluate in simulation; however, note that it may not perfectly correlate with human judgments of fold quality (8).

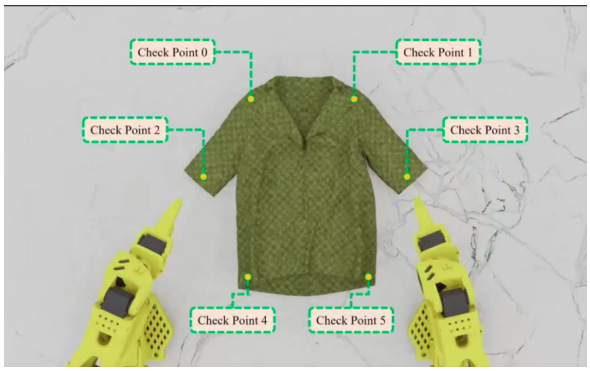

Figure 3: Quantitative evaluation metric

3.1 Quantitative Performance

Quantitative evaluations showed that SmolVLA outperformed the other policies with a success rate of 69%. ACT followed with a 61.11% success rate, while the Diffusion Policy reached only 41% success. In qualitative visual assessments, the Diffusion Policy was noted as the least effective, whereas SmolVLA and ACT demonstrated high performance.

4 Discussion and Insights

Several key learnings emerged during the development of our cloth-folding pipeline. We categorize these insights across five areas:

- **Inference Challenges.** Smooth simulation in Isaac Sim was best achieved using CPU resources rather than GPU, which was counter-intuitively to our initial assumption. It was able to significantly increase the smoothness and speed of the simulation during policy rollouts, reducing render time from 40 minutes per frame to near real-time playback.

- **Teleoperation Challenges.** Alternatively for collecting high-quality demonstrations we noticed a significant amount of GPU was required to run Isaac Sim for data collection. Latency in the leader/follower SO101 arm setup was a recurring bottleneck during our attempts at data collection.

- **Diversity vs. Quantity.** The diversity of initial cloth configurations has proven more critical to policy performance than the sheer volume of demonstrations. Policies trained on a broader distribution of starting states generalize substantially better, motivating our use of Cosmos-Transfer 2.5 augmentation to artificially expand state coverage for our next iteration on dataset.

- **Reachability Problems.** All three policies struggled to learn adaptive behavior when target grasp points fell outside the robot's reachable workspace. This failure mode was particularly prevalent for heavily crumpled garments, and resolving it will require dedicated training data capturing edge-case configurations alongside learned workspace-aware grasp selection.

- **Failure Recovery.** A significant limitation observed across all policies was the absence of recovery behaviors: once an initial trajectory deviated from the demonstrated path, the policy continued executing the original plan rather than re-grasping or re-planning. Incorporating corrective intervention data or online

Episode	Garment	SmolVLA	DP	ACT
1	Top_Long_Seen_0	3/5	2/5	2/5
2	Top_Long_Seen_0	3/5	2/5	3/5
3	Top_Long_Seen_1	5/5	4/5	5/5
4	Top_Long_Seen_1	5/5	4/5	5/5
5	Top_Long_Seen_2	4/5	2/5	2/5
6	Top_Long_Seen_2	5/5	1/5	5/5
Success Rate		**69%**	**41%**	**61%**

Table 1: Number of passing keypoint pairs (out of 5 total) when folding a set of 3 longsleeve shirts in simulation (with 2 episodes per shirt) using each of the three policies. SmolVLA and ACT both perform relatively well, while Diffusion Policy lags significantly behind.

RL fine-tuning is a natural next step to address this brittleness that we plan to incorporate before our final submission. This recovery failure was most noticeable when evaluating diffusion policy.

Taken together, these findings suggest that future work should prioritize curriculum-based data collection strategies, workspace-conditioned grasp planning, and recovery-aware policy objectives over simply scaling demonstration counts. Furthermore, over the coming weeks we will be implementing automated trajectory generation to scale our dataset beyond manual teleoperation. We will model the S101 arms in MagicSim to procedurally generate automated trajectories, enabling the high-volume synthesis of physics-verified folding sequences and error-recovery behaviors.

5 Conclusion and Future Work

In this work, we validated a foundational framework for bimanual garment manipulation, demonstrating that the SmolVLA architecture achieves a superior success rate of 69% compared to ACT and Diffusion Policy baselines. These results represent the baseline for our LeHome Challenge 2026 entry.

Over the next five weeks leading up to the simulation submission deadline, we will focus on significantly increasing our evaluation metrics by incorporating automated trajectory generation via **MagicSim**. This will allow us to procedurally synthesize physics-verified recovery behaviors and address current reachability limitations. Following the simulation evaluation phase, we will advance our sim-to-real pipeline through extensive domain randomization and real-world data collection to ensure robust performance for the ICRA 2026 on-site finals.

Acknowledgments

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